

Roll No.....

END SEMESTER EXAMINATION DEC - 2025

Name of the Program: B. Tech Petroleum Engineering
Name of the Course: Petroleum Drilling Engineering II

Semester: 5th
Course Code: TPE 504

Time: 3 Hrs

Maximum Marks: 100

Notes:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a,b,c in each main question.
- (iii) Total marks for each main question is twenty. Each questions carries 10 marks

Q1	(2X10 = 20 marks)	CO
(a)	Define directional drilling and deviation control. Describe the applications of directional drilling.	CO1
(b)	What are the parameters to be considered in direction well planning?	CO1
(c)	It is desired to drill under the lake to the location designated for Well 2. For this well build-and-hold trajectory will be used Horizontal departure to the target is 2655 ft at TVD of 9650 ft. The recommended rate of build is 2.0°/100 ft. The kickoff depth is 1600 ft. Determine (b1) the radius of curvature R1 (b2) the maximum inclination angle θ (b3) the measured depth to the end of build (b4) the total measured depth	CO6
Q2	(2X10 = 20 marks)	CO
(a)	What is a multilateral well? List and describe the flow control devices used in multilateral wells and discuss their applications.	CO2
(b)	Calculate Torque while rotating off bottom in the horizontal well using following well data: Build curve radius (R) = 1042 ft Horizontal section = 1500 ft WOB = 30,000 lbf Mud weight = 9.2 ppg BHA is made up mostly of HWDP HWDP = 5", 50 lb/ft, OD of tool joint is 6.5"	CO6
(c)	Describe cased-hole and whipstock kick-offs used in multilateral wells, including their advantages and disadvantages.	CO2
Q3	(2X10 = 20 marks)	CO
(a)	Explain how to free a differentially stuck pipe? List the preventive measures for avoiding differential sticking.	CO3
(b)	Explain what is lost circulation problem during drilling of a well? What preventive measures can be taken to minimize lost circulation?	CO3
(c)	List and describe the key operational techniques to prevent hole cleaning problems in deviated or horizontal wells.	CO3
Q4	(2X10 = 20 marks)	CO
(a)	A high-pressure, high-temperature (HPHT) well is planned. Discuss the key casing design considerations.	CO4

(b)	What are the factors considered when selecting casing size, weight, and grade?	CO4
(c)	Discuss the collapse and burst criterion for casing design.	CO4
Q5	(2X10 = 20 marks)	
(a)	Discuss the role of material properties (tensile strength, yield strength, toughness) in drill pipe selection	CO5
(b)	A drill pipe is subjected to a maximum tensile load of 1,200 kN. Calculate the minimum required cross-sectional area if the material has a yield strength of 690 MPa.	CO5
(c)	Explain the key factors to consider when choosing a BHA for directional drilling."	CO5